

# Managing Users in MySQL

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## What is a User?

A **user** in MySQL is an account that can connect to the database. Each user has a **username**, **host**, and **password**.

Example:

```
'user1'@'localhost'
```

- **user1** → Username
  - **localhost** → Host from which the user can connect
- 

## 1. Create a User

Used to create a new MySQL user.

**Syntax:**

```
CREATE USER 'username'@'host' IDENTIFIED BY 'password';
```

**Example:**

```
CREATE USER 'john'@'localhost' IDENTIFIED BY 'john123';
```

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## 2. View Existing Users

To view all MySQL users:

```
SELECT User, Host FROM mysql.user;
```

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## 3. Change User Password

### Syntax:

```
ALTER USER 'username'@'host'  
IDENTIFIED BY 'new_password';
```

### Example:

```
ALTER USER 'john'@'localhost'  
IDENTIFIED BY 'newpass123';
```

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## 4. Rename a User

### Syntax:

```
RENAME USER 'old_user'@'host'  
TO 'new_user'@'host';
```

### Example:

```
RENAME USER 'john'@'localhost'  
TO 'john_doe'@'localhost';
```

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## 5. Delete a User

Removes a user account permanently.

### Syntax:

```
DROP USER 'username'@'host';
```

### Example:

```
DROP USER 'john'@'localhost';
```

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## 6. Grant Privileges

Gives permissions to a user.

### Syntax:

```
GRANT privilege
ON database.table
TO 'username'@'host';
```

### Example:

```
GRANT SELECT, INSERT
ON school.*
TO 'john'@'localhost';
```

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## 7. View User Privileges

Displays all privileges assigned to a user.

```
SHOW GRANTS FOR 'john'@'localhost';
```

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## 8. Revoke Privileges

Removes previously granted permissions.

### Syntax:

```
REVOKE privilege
ON database.table
FROM 'username'@'host';
```

### Example:

```
REVOKE INSERT
ON school.*
FROM 'john'@'localhost';
```

## 9. Flush Privileges (Optional)

Reloads the privilege tables after manual changes.

```
FLUSH PRIVILEGES;
```

Usually **not required** after `CREATE USER` , `GRANT` , `REVOKE` , or `ALTER USER` , as these commands update privileges automatically.

## Common Privileges

| Privilege      | Purpose                 |
|----------------|-------------------------|
| SELECT         | Read data               |
| INSERT         | Add new records         |
| UPDATE         | Modify records          |
| DELETE         | Remove records          |
| CREATE         | Create databases/tables |
| DROP           | Delete databases/tables |
| ALTER          | Modify table structure  |
| ALL PRIVILEGES | Grant all permissions   |

## Summary

| Command     | Purpose                             |
|-------------|-------------------------------------|
| CREATE USER | Create a new user                   |
| ALTER USER  | Change password or account settings |
| RENAME USER | Rename a user                       |
| DROP USER   | Delete a user                       |

| Command          | Purpose                                 |
|------------------|-----------------------------------------|
| GRANT            | Give privileges                         |
| SHOW GRANTS      | View privileges                         |
| REVOKE           | Remove privileges                       |
| FLUSH PRIVILEGES | Reload privilege tables (rarely needed) |

**Remember:** A MySQL user is identified by both **username** and **host** (for example, `'john'@'localhost'` ).